

# Problem Set 1

This problem set is meant to help you familiarize yourself with Python and Pandas. This is meant to be a very gentle introduction -- if you find this problem set to be difficult, you might want to consider taking a more introductory course.

## Before You Start

For this problem set, you should download INF0251-PS1.ipynb from bCourses. Create a local copy of the notebook and rename it LASTNAME\_FIRSTNAME-PS1.ipynb. Then edit your renamed file directly in your browser by typing:

```
jupyter notebook <name_of_downloaded_file>
```

Make sure the following libraries load correctly (hit Ctrl-Enter).

```
In [ ]: #IPython is what you are using now to run the notebook
import IPython
print("IPython version:      %6.6s (need at least 1.0)" % IPython.__version__)

# Numpy is a library for working with Arrays
import numpy as np
print("Numpy version:      %6.6s (need at least 1.7.1)" % np.__version__)

# SciPy implements many different numerical algorithms
import scipy as sp
print("SciPy version:      %6.6s (need at least 0.12.0)" % sp.__version__)

# Pandas makes working with data tables easier
import pandas as pd
print("Pandas version:      %6.6s (need at least 0.11.0)" % pd.__version__)

# Module for plotting
import matplotlib
print("Matplotlib version:   %6.6s (need at least 1.2.1)" % matplotlib.__version__)

# SciKit Learn implements several Machine Learning algorithms
import sklearn
print("Scikit-Learn version: %6.6s (need at least 0.13.1)" % sklearn.__version__)

IPython version:      8.5.0 (need at least 1.0)
Numpy version:        1.24.2 (need at least 1.7.1)
SciPy version:        1.10.1 (need at least 0.12.0)
Pandas version:       1.5.3 (need at least 0.11.0)
Matplotlib version:   3.7.1 (need at least 1.2.1)
Scikit-Learn version: 1.4.0 (need at least 0.13.1)
```

## Working in a group?

List the names of other students with whom you worked on this problem set:

- None
- 

## Introduction to the assignment

For this assignment, you will be using the [California Housing Prices Dataset](#). Please read about the dataset carefully before continuing -- it is worth investing a few minutes up front otherwise you are likely to be hopelessly confused! We'll be coming back to this dataset repeatedly throughout the semester. Also, if you're new to analyzing data in Python, please make sure to read the relevant readings linked to on Canvas before beginning, otherwise you'll be stabbing in the dark.

Use the following commands to load the dataset:

```
In [ ]: from sklearn.datasets import fetch_california_housing  
cal_data = fetch_california_housing()
```

The following commands will provide some basic information about the data:

```
In [ ]: print(cal_data.DESCR)  
print(cal_data.keys())  
print(cal_data.feature_names)  
print(cal_data.data.shape)  
print(cal_data.target.shape)
```

```
.. _california_housing_dataset:
```

California Housing dataset

---

**\*\*Data Set Characteristics:\*\***

:Number of Instances: 20640

:Number of Attributes: 8 numeric, predictive attributes and the target

:Attribute Information:

- MedInc            median income in block group
- HouseAge        median house age in block group
- AveRooms        average number of rooms per household
- AveBedrms       average number of bedrooms per household
- Population      block group population
- AveOccup        average number of household members
- Latitude        block group latitude
- Longitude       block group longitude

:Missing Attribute Values: None

This dataset was obtained from the StatLib repository.  
[https://www.dcc.fc.up.pt/~ltorgo/Regression/cal\\_housing.html](https://www.dcc.fc.up.pt/~ltorgo/Regression/cal_housing.html)

The target variable is the median house value for California districts, expressed in hundreds of thousands of dollars (\$100,000).

This dataset was derived from the 1990 U.S. census, using one row per census block group. A block group is the smallest geographical unit for which the U. S.

Census Bureau publishes sample data (a block group typically has a population of 600 to 3,000 people).

A household is a group of people residing within a home. Since the average number of rooms and bedrooms in this dataset are provided per household, these columns may take surprisingly large values for block groups with few households and many empty houses, such as vacation resorts.

It can be downloaded/loaded using the  
 :func:`sklearn.datasets.fetch\_california\_housing` function.

.. topic:: References

- Pace, R. Kelley and Ronald Barry, Sparse Spatial Autoregressions, Statistics and Probability Letters, 33 (1997) 291-297

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude']
(20640, 8)
(20640,)
```

The following commands will put together the features and target into a pandas dataframe:

```
In [ ]: cal_df = pd.DataFrame(
        data=cal_data.data,
        columns=cal_data.feature_names
    )
cal_df['MedHouseVal'] = cal_data['target']
```

## Part 1: Descriptive analysis

### 1.1: Explore the data

Let's dig into the data a bit to see what we're dealing with. The first thing to do is to make sure you understand how the data is organized, what the data types are, whether there is any missing data, and so forth. Get your bearings on your own, then answer the following questions.

- 1.1.1: How many different variables are there in the dataset, and how many different observations?
- 1.1.2: What is the datatype of each variable?
- 1.1.3: Does the dataset contain any missing values?
- 1.1.4: How would you interpret the row index value?

```
In [ ]: cal_df
```

```
Out[ ]:
```

|       | MedInc | HouseAge | AveRooms | AveBedrms | Population | AveOccup | Latitude | Longitude |
|-------|--------|----------|----------|-----------|------------|----------|----------|-----------|
| 0     | 8.3252 | 41.0     | 6.984127 | 1.023810  | 322.0      | 2.555556 | 37.88    | -122.23   |
| 1     | 8.3014 | 21.0     | 6.238137 | 0.971880  | 2401.0     | 2.109842 | 37.86    | -122.22   |
| 2     | 7.2574 | 52.0     | 8.288136 | 1.073446  | 496.0      | 2.802260 | 37.85    | -122.24   |
| 3     | 5.6431 | 52.0     | 5.817352 | 1.073059  | 558.0      | 2.547945 | 37.85    | -122.25   |
| 4     | 3.8462 | 52.0     | 6.281853 | 1.081081  | 565.0      | 2.181467 | 37.85    | -122.25   |
| ...   | ...    | ...      | ...      | ...       | ...        | ...      | ...      | ...       |
| 20635 | 1.5603 | 25.0     | 5.045455 | 1.133333  | 845.0      | 2.560606 | 39.48    | -121.09   |
| 20636 | 2.5568 | 18.0     | 6.114035 | 1.315789  | 356.0      | 3.122807 | 39.49    | -121.21   |
| 20637 | 1.7000 | 17.0     | 5.205543 | 1.120092  | 1007.0     | 2.325635 | 39.43    | -121.22   |
| 20638 | 1.8672 | 18.0     | 5.329513 | 1.171920  | 741.0      | 2.123209 | 39.43    | -121.32   |
| 20639 | 2.3886 | 16.0     | 5.254717 | 1.162264  | 1387.0     | 2.616981 | 39.37    | -121.24   |

20640 rows x 9 columns

```
In [ ]: # Your code here

# 1.1 data variables:
print(len(cal_df.columns))
```

```
# 1.1 observations
print(len(cal_df))

# 1.2 datatype
print(cal_df.dtypes)

# 1.1.3 missing values
print(cal_df.isnull().values.any())
```

```
9
20640
MedInc          float64
HouseAge        float64
AveRooms        float64
AveBedrms       float64
Population      float64
AveOccup        float64
Latitude        float64
Longitude       float64
MedHouseVal     float64
dtype: object
False
False
```

*Provide your answers below:*

- 1.1.1: There are 8 predictive variables and a single target variable with 20640 observations in each variable
- 1.1.2: Each variable is float64 type.
- 1.1.3: There are no missing values.
- 1.1.4: The row index value corresponds to (n-1)th census block group in California

## 1.2: Answer some basic questions

- 1.2.1: What is the average population per block group? What was California's total population in 1990 according to the Census?
- 1.2.2: What are the median house values in California's block groups with the lowest and highest populations?
- 1.2.3: Suggest 5 block groups that are likely to be vacation resorts. Do a quick Google search to validate your results.
- 1.2.4: How many census block groups are west of the city of Berkeley (lat: 37.871666, long: -122.272781)?
- 1.2.5: What fraction of block groups have an average number of household members greater or equal to 10?

```
In [ ]: # Your code here
# 1.2.1
print(cal_df['Population'].mean())
print(cal_df['Population'].sum())

#1.2.2
```

```

print(cal_df['MedHouseVal'][cal_df['Population'].idxmax()]) # max pop house val
print(cal_df['MedHouseVal'][cal_df['Population'].idxmin()]) # min pop house val

#1.2.3
print(cal_df.sort_values('AveOccup', ascending =True).head())

#1.2.4
print(len(cal_df[cal_df['Longitude']< -122.272781])) # west is more negative lon

#1.2.5
print(len(cal_df[cal_df['AveOccup'] >= 10])/ len(cal_df))

```

1425.4767441860465

29421840.0

1.344

3.5

|       | MedInc | HouseAge | AveRooms  | AveBedrms | Population | AveOccup | Latitude |
|-------|--------|----------|-----------|-----------|------------|----------|----------|
| 11849 | 2.5893 | 8.0      | 28.615385 | 5.487179  | 27.0       | 0.692308 | 39.75    |
| 5342  | 0.5360 | 16.0     | 4.500000  | 1.500000  | 3.0        | 0.750000 | 34.04    |
| 19522 | 0.6825 | 17.0     | 2.372549  | 0.990196  | 198.0      | 0.970588 | 37.65    |
| 19427 | 2.2500 | 52.0     | 3.090909  | 0.696970  | 35.0       | 1.060606 | 37.71    |
| 4623  | 0.8804 | 36.0     | 2.713235  | 1.080882  | 145.0      | 1.066176 | 34.06    |

|                       | Longitude | MedHouseVal |
|-----------------------|-----------|-------------|
| 11849                 | -121.00   | 0.830       |
| 5342                  | -118.44   | 3.500       |
| 19522                 | -121.00   | 1.875       |
| 19427                 | -121.00   | 1.750       |
| 4623                  | -118.31   | 4.500       |
| 2167                  |           |             |
| 0.0017926356589147287 |           |             |

Provide your answers below:

- 1.2.1: The average population per block group is 1425. The total population in 1990 was 29421840 according to the census.
- 1.2.2: The median house values in California's block groups with the lowest population is 3, 500, 000 and the highest population is 1,344,000.
- 1.2.3: Block groups that have the lowest Average Occupancy are likely to be empty houses such as vacation resorts. Sorting the data by ascending average occupancy we obtain blocks 11850, 5343, 19523, 19428 and 4624. After doing a quick google search indeed the 2nd census block is located in Beverly Hills while the 5th one is Garden Suite Hotel.
- 1.2.4: There are 2167 census block groups west of the city of Berkeley (more negative longitude. Note: we are ignoring latitude here, can be NW or SW also)
- 1.2.5: The fraction of block groups that have an average number of household members greater or equal to 10 is .00179

## 1.3: Summary statistics

Create a clean, organized table that shows just the following information (no more, no less) for each variable in the dataset. Note that your table should have K rows (one for each variable) and 7 columns, ordered as below:

- The name of the variable
- The number of observations with non-missing values
- The mean of the variable
- The standard deviation of the variable
- The minimum value of the variable
- The median of the variable
- The maximum value of the variable

```
In [ ]: # enter your code here
cal_summary = pd.DataFrame(columns = ['VarName', 'ObservationNum', 'Mean', 'StdDev', 'Min', 'Median', 'Max'])
cal_summary['VarName'] = list(cal_df.columns)
cal_summary['ObservationNum'] = cal_df.notnull().sum(axis=0).values
cal_summary['Mean'] = cal_df.mean(axis=0).values
cal_summary['Min'] = cal_df.min(axis=0).values
cal_summary['Median'] = cal_df.median(axis=0).values
cal_summary['Max'] = cal_df.max(axis=0).values
cal_summary['StdDev'] = cal_df.std(axis=0).values
cal_summary.drop(index=8, inplace = True) # drop target variable since we have it
print(cal_summary)
```

|   | VarName    | ObservationNum | Mean        | StdDev      | Min         | \ |
|---|------------|----------------|-------------|-------------|-------------|---|
| 0 | MedInc     | 20640          | 3.870671    | 1.899822    | 0.499900    |   |
| 1 | HouseAge   | 20640          | 28.639486   | 12.585558   | 1.000000    |   |
| 2 | AveRooms   | 20640          | 5.429000    | 2.474173    | 0.846154    |   |
| 3 | AveBedrms  | 20640          | 1.096675    | 0.473911    | 0.333333    |   |
| 4 | Population | 20640          | 1425.476744 | 1132.462122 | 3.000000    |   |
| 5 | AveOccup   | 20640          | 3.070655    | 10.386050   | 0.692308    |   |
| 6 | Latitude   | 20640          | 35.631861   | 2.135952    | 32.540000   |   |
| 7 | Longitude  | 20640          | -119.569704 | 2.003532    | -124.350000 |   |

|   | Median      | Max          |
|---|-------------|--------------|
| 0 | 3.534800    | 15.000100    |
| 1 | 29.000000   | 52.000000    |
| 2 | 5.229129    | 141.909091   |
| 3 | 1.048780    | 34.066667    |
| 4 | 1166.000000 | 35682.000000 |
| 5 | 2.818116    | 1243.333333  |
| 6 | 34.260000   | 41.950000    |
| 7 | -118.490000 | -114.310000  |

## 1.4 Simple Linear Regression

Estimate a linear regression of the median house value (the dependent variable) on the population (the independent variable), with no other control variables. Interpret the coefficients and standard errors. Based on this analysis, can you conclude anything about the causal effect of decreasing the population on the median housing value?

```
In [ ]: import statsmodels.api as sm

# define the independent and dependent variables
X = cal_df['Population']
y = cal_df['MedHouseVal']

# Add a constant term - ensures we have a constant term in the model
X = sm.add_constant(X)

# fit regression model
model = sm.OLS(y, X).fit()

# print the summary of OLS regression results
print(model.summary())
```

### OLS Regression Results

```
=====
Dep. Variable:          MedHouseVal      R-squared:                0.001
Model:                  OLS              Adj. R-squared:           0.001
Method:                 Least Squares     F-statistic:              12.55
Date:                   Mon, 22 Jan 2024   Prob (F-statistic):       0.000398
Time:                   21:24:41          Log-Likelihood:           -32236.
No. Observations:       20640            AIC:                     6.448e+04
Df Residuals:           20638            BIC:                     6.449e+04
Df Model:                1
Covariance Type:        nonrobust
=====
```

|            | coef       | std err  | t       | P> t  | [0.025   | 0.975]    |
|------------|------------|----------|---------|-------|----------|-----------|
| const      | 2.1044     | 0.013    | 163.012 | 0.000 | 2.079    | 2.130     |
| Population | -2.512e-05 | 7.09e-06 | -3.542  | 0.000 | -3.9e-05 | -1.12e-05 |

```
=====
Omnibus:                2387.069      Durbin-Watson:           0.308
Prob(Omnibus):           0.000        Jarque-Bera (JB):        3301.867
Skew:                   0.967         Prob(JB):                0.00
Kurtosis:               3.311         Cond. No.:               2.93e+03
=====
```

#### Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.93e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Write your response here.

- The intercept is 210,436 dollars as the median house value for a block of population of zero, while the slope is -2.512e-05 dollars per person. Since the slope is negative it indicates a trend to have higher median house values in smaller population blocks. The standard error is large for the constant parameter indicating there is high dispersion in the data set. The slope has a smaller standard error but it is still large relative to the absolute value of slope.

- Although there appears to be a correlation between the median house value and the size of the population block, we cannot conclude any causal effect of decreasing the population on the median housing value since the standard errors are non-negligible.

## Part 2: Histograms and Scatterplots

### 2.1: Histogram of housing prices

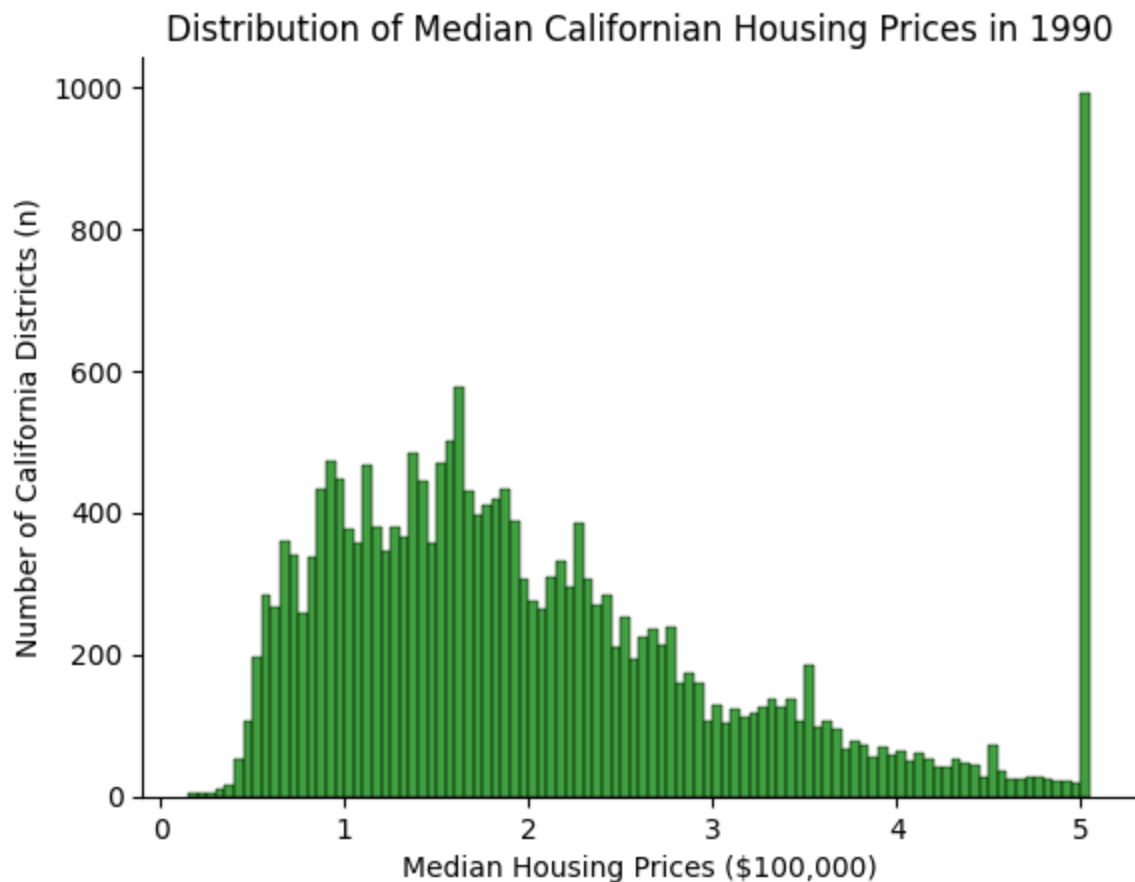
Below you will find some very basic code to make a very basic histogram of median housing prices (the "target" variable) for your dataset. Your first task is to make this graph look pretty by doing the following:

1. Add appropriate labels to the x and y axes, make sure to include units
2. Change the bin width on the histogram to be \$5,000
3. Remove the axes on the top and right side of the plot window
4. Change the color of the bars to be green
5. Add an appropriate title

```
In [ ]: # prepare IPython to work with matplotlib and import the library to something c
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns

# edit the code below to make the graph look good
sns.histplot(cal_df['MedHouseVal'], color = 'green', binwidth = 0.05) #2.1.2 and
plt.xlabel('Median Housing Prices ($100,000)') # 2.1.1
plt.ylabel('Number of California Districts (n)') #2.1.1
sns.despine() #2.1.3
plt.title('Distribution of Median Californian Housing Prices in 1990') #2.1.5
```

```
Out[ ]: Text(0.5, 1.0, 'Distribution of Median Californian Housing Prices in 1990')
```

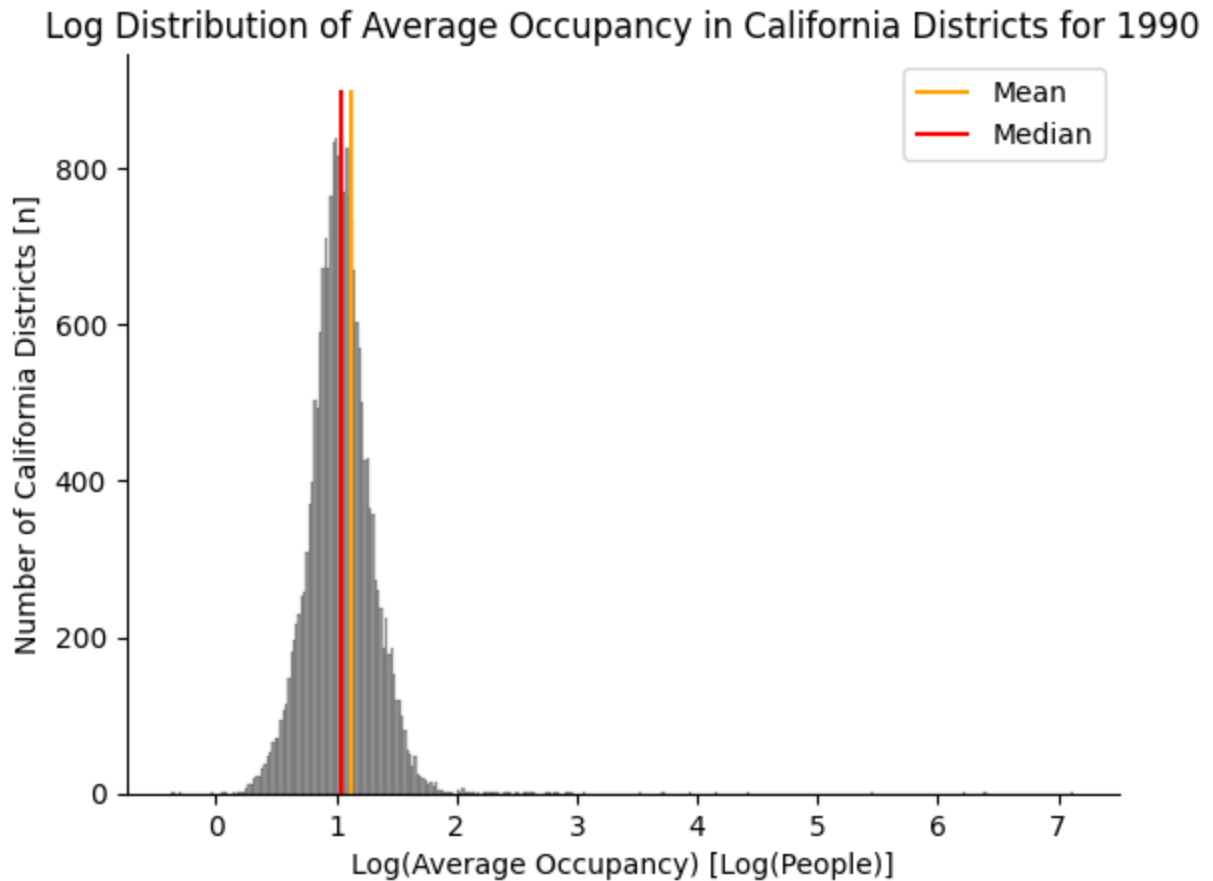


## 2.2: Histogram of average occupancy

Now use your histogramming skills to create a fine looking histogram of the average number of household members ("AveOccup"). In the same figure, plot the mean and median values of this variable. (Hint: applying a very common transformation to the data might make things easier).

```
In [ ]: # enter your code here
sns.histplot(np.log(cal_df['AveOccup']), color = 'grey')
plt.xlabel('Log(Average Occupancy) [Log(People)]')
plt.ylabel('Number of California Districts [n]')
sns.despine()
plt.vlines(np.log(cal_df['AveOccup'].mean()), 0, 900, color = 'orange', label = 'Mean')
plt.vlines(np.log(cal_df['AveOccup'].median()), 0, 900, color = 'red', label = 'Median')
plt.title('Log Distribution of Average Occupancy in California Districts for 1990')
plt.legend()
```

```
Out[ ]: <matplotlib.legend.Legend at 0x2bbd3efe0>
```



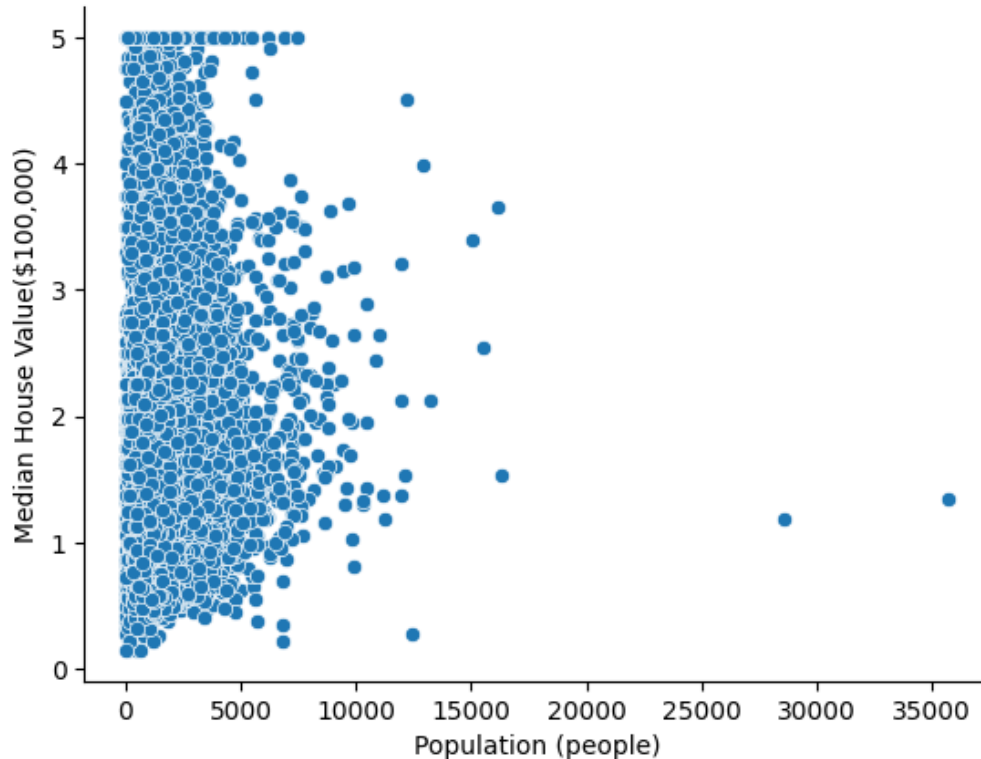
## 2.3: Scatter plot of housing prices and population

Use matplotlib to create a scatter plot that shows the relationship between the block group median house value (y-axis) and the block group population (x-axis). Properly label your axes, and make sure that your graphic looks polished and professional.

```
In [ ]: # enter your code here
sns.scatterplot(x = cal_df['Population'], y = cal_df['MedHouseVal'])
sns.despine()
plt.ylabel('Median House Value($100,000)')
plt.xlabel('Population (people)')
plt.title(' Median house value as a function of population for Californian dist
```

```
Out[ ]: Text(0.5, 1.0, ' Median house value as a function of population for California
n districts in 1990')
```

## Median house value as a function of population for Californian districts in 1990



### 2.4: Interpret

What do you observe in the above scatter plot? Does there appear to be a relationship between media house value and population in California? Calculate the correlation between these two variables. Do you think this relationship is causal, or just a correlation? Justify your position and compare to your answer in section 1.4.

```
In [ ]: # enter your code here

correlation = cal_df['Population'].corr(cal_df['MedHouseVal'])
print("Correlation coefficient:", correlation)
```

Correlation coefficient: -0.024649678888894896

Write your thoughts here (1-2 paragraphs).

- From the scatterplot above we can see that the majority of the house values are heavily concentrated in a single region of the graph spanning from 0 - 5000 people per block. The median house values in this portion of the graph are distributed equally across 0 - 500k\$. This demonstrates that there is likely little correlation between the the median house value and population in california, as you get past 5000 people into 35000 people you begin to see that there might be a slight negative correlation between population and housing value, but this is not very strong.
- We can caluclate the correlation (pearson r value) between these two variables which gives us -0.02464 which is not a very strong relationship. Therefore, we can say the relationship betwwen median housing value and population is merely a weak negative correlation. These results are consistent from the linear regression we performed in 1.4

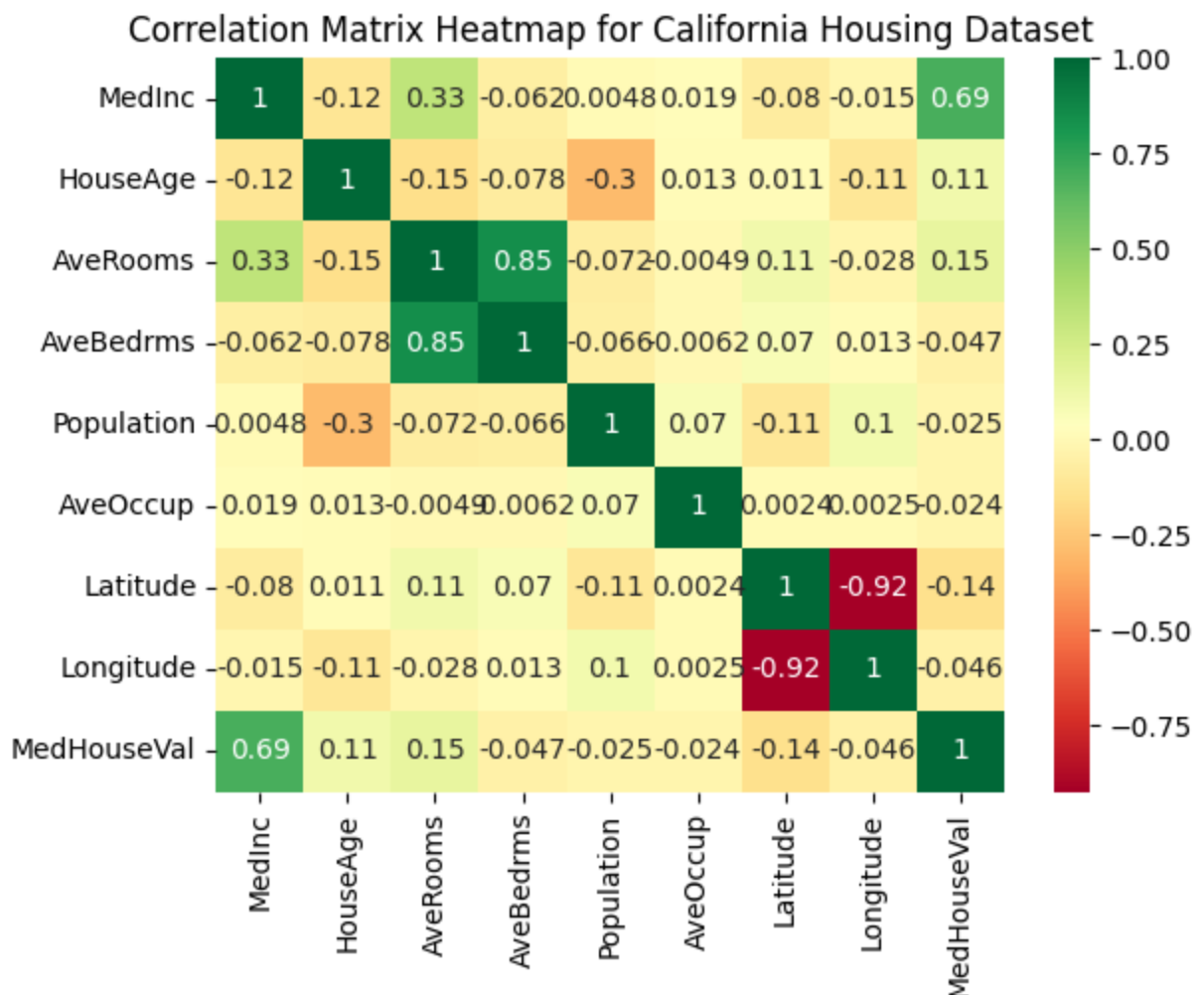
where we found the standard error to be quite large for both of the parameters in our model. This suggests a weak correlation and not a very strong relationship between this variable and our target. Therefore our statement that we cannot make a causal inference here is correct.

## 2.5 Correlation Matrix

Calculate the correlation of every pair of variables in the dataset. Create a  $K \times K$  matrix where the value in the  $(i,j)$  cell is the correlation between the  $i$ th and  $j$ th variable. Show off your skills by coloring the cell so that large positive correlations appear green and large negative correlations appear red (use a gradient to make this pretty). What two variables appear to me most positively and negatively correlated? Explain these results.

```
In [ ]: # enter your code here
correlation_matrix = cal_df.corr() # including the target here
# print(correlation_matrix)

# Visualize
sns.heatmap(correlation_matrix, annot=True, cmap="RdYlGn")
plt.title("Correlation Matrix Heatmap for California Housing Dataset")
plt.show()
```



Write your response here.

- The variables that appear to be the most positively correlated are Median house value and median household income. This makes sense since its likley households with larger income can afford a more expensive home. The variables with largest negative correlation were longitude and latitude which makes sense given the geographical location of California that its located in -120s longitude and +30s latitude and confined to a relatively small region of earth. These will tend to be anticorrelated always.

## 2.6 Create your own (creative and effective) visualization

Use another type of graph or chart to illustrate an interesting pattern in the data. Be creative in your visualization, and make sure to produce a "publication-quality" graph. Points will be given for useful and creative graphs; points will be deducted for confusing issues like unlabeled axes. If you're new to data visualization, [this guide](#) is a good place to start.

```
In [ ]: # enter your code here
sns.scatterplot(x=cal_df['Longitude'], y=cal_df['Latitude'], size=cal_df['MedInc'],
               style=cal_df['MedHouseVal'])
sns.despine()
plt.legend(bbox_to_anchor=(1.01, 0.99), loc="upper left")
plt.title("Median house value and median income as a function of district location in California")
```

```
Out[ ]: Text(0.5, 1.0, 'Median house value and median income as a function of district location in California')
```

Median house value and median income as a function of district location in California

